Wall time per step: Parse 1M PBMC, full (684,932 cells)

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells	CyteArc mtx	CyteArc h5ad
load	98	48	43	41	OOM	OOM	40	OOM	OOM	731	OOM	OOM	142	384	143
qc	3.1	3.2	2.1	2.0	–	–	7.4	–	–	13	–	–	9.4	0.1	0.1
normalise	n/a	n/a	n/a	n/a	–	–	0.0	–	–	0.0	–	–	4.6	27	28
hvg	2.0	2.0	1.7	1.6	–	–	8.5	–	–	27	–	–	27	33	32
pca	4.3	4.0	2.1	1.7	–	–	17	–	–	106	–	–	696	38	41
harmony	3.7	3.8	2.3	1.1	–	–	4.5	–	–	255	–	–	217	228	240
knn_snn	24	25	13	12	–	–	33	–	–	61	–	–	70	167	172
cluster	8.7	8.8	6.9	6.6	–	–	30	–	–	217	–	–	256	17	17
cluster_fast	37	37	4.8	3.9	n/a	n/a	n/a	–	–	330	n/a	n/a	n/a	n/a	n/a
umap	52	51	5.8	5.2	–	–	349	–	–	262	–	–	454	588	588
dge	5.1	5.3	4.6	4.3	–	–	–	–	–	TIMEOUT	–	–	1,137	39	40
tsne_default	372	376	323	321	–	–	1,941	n/a	n/a	n/a	–	–	1,606	n/a	n/a
tsne_fft	116	115	20	31	–	–	247	n/a	n/a	n/a	–	–	232	n/a	n/a
tsne_quick	82	83	41	42	n/a	n/a	n/a	–	–	–	n/a	n/a	n/a	n/a	n/a
total (no tSNE)	202	151	81	75	failed	failed	490	failed	failed	failed	failed	failed	3,012	1,521	1,300

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.